



The Effect of Welding Speed on the Mechanical Properties of the FSW Cu/Al

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Abstract. This study was carried out to join pure copper C1100 plate and aluminum 1050 plate by friction stir welding. The effect of the welding speed on the macrostructure and mechanical properties of the joints was investigated. At various welding speeds, the tunnel defects seemed to be inevitable. However, at high speed, the tunnel size is small, and the size is increased with the reduction of the tool offset. The tensile strength reached the highest value at a welding regime with the tool offset around 2.5 mm, and the fractures took place at the interface. The strength bonding between copper and aluminum could be obtained at least 80% of the base aluminum 1050.

Keywords: Dissimilar joint · FSW · Defects · Tensile strength

1 Introduction

Friction Stir Welding (FSW) has been emerging as a green technique in metal joining with high energy efficiency, environment friendliness, and versatility. This technique is suitable for joining the dissimilar metals that are un-weldable by traditional fusion welding methods. The basic concept of FSW process is quite simple. The rotating shoulder and probe of a non-consumable tool heat and plasticize the surrounding metal, and a solid-state joint is established.

Recently, with the development of electric vehicles and equipment, several researchers have addressed to join the dissimilar electrical materials, i.e., copper to aluminum. Since the significant different properties between copper and aluminum alloy, the bonding between these two metals becomes a big challenge. Several researched works [1–5] have been conducted to join these metals by friction stir welding technique. However, the microstructure and mechanical properties of the bonding was quite sensitive to the welding parameters. The non-uniform materials in the welded zone during the stirring process must be one of the main culprits. In this work, we try to fabricate the FSW joint between pure copper C1100 and aluminum

1050 in the Friction welding Lab of Nha Trang University. The effect of the welding speed on the defects and mechanical properties of the joint was carried out.

2 Experimental Procedures

The pure copper C1100 plate and aluminum 1050 plate with 5.0 mm thickness were butt-joined by friction stir welding using the CNC milling machine. A cylindrical tool with a pin diameter of 8.0 mm and 4.8 mm length was used. The pin is aligned at a tilt angle of 2.0° in the plane containing pin axis and center weld line (the tilt angle is defined as the angle between pin axis and the direction perpendicular to the work-pieces). Several joints were fabricated at a constant rotation tool 800 rpm with various welding speeds (from 70 to 200 mm/min) (Fig. 1).

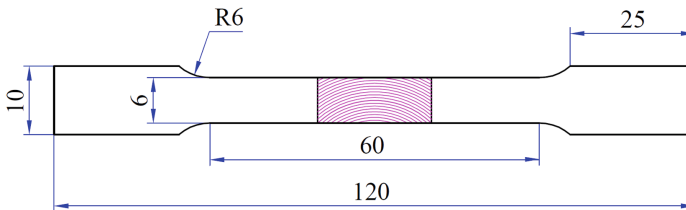


Fig. 1. Geometry of tensile specimen.

The microstructure was observed by an Olympus microscope. The tensile specimens were machined from the welded plate with a thickness of 5.0 mm and the geometry based on ASTM E08. The tensile specimen loading direction was perpendicular to the welding line. The tensile tests were performed by the Instron machine 3366, 10 kN at a constant strain rate of 5 mm/min.

3 Results and Discussion

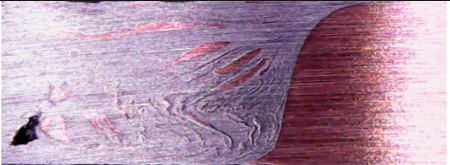
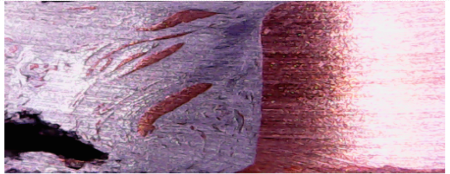
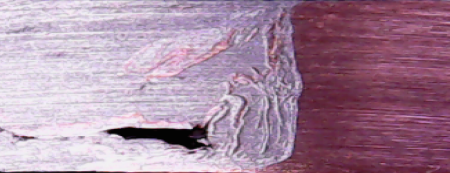
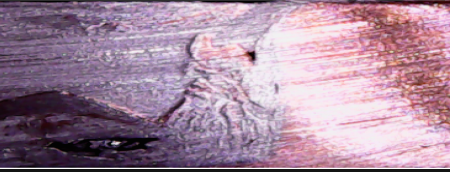
The macrostructure of the joints at various welding speeds was summarized in Table 1. In these macrostructures in Table 1, the aluminum is on the left side, and copper is on the right side.

The bonding between copper 1100 and aluminum 1050 could be obtained advantageously by the friction stir welding process. The bonding strength could be gained up to 85 MPa ($\approx 67\%$ tensile strength of base aluminum 1050). However, the tunnel defects seemed to be dominant in the joints, see Table 1. In all welding regimes, the copper material that was stirred into the aluminum can be obviously seen in Table 1 and Fig. 2. It should be noted that the strength efficiency of the joint between A1050 and Cu1100 could be obtained at 80% [6]. Thus many efforts must be done to improve this joint in the near future.

The Fig. 3 shows the tensile strength of the joint at various welding speeds. The strength of the joint seems to reduce with the increasing of the welding speed.

The strength of the joint reaches about 85 MPa with the welding speed below 100 mm/min. Then, with increasing the welding speed, the tensile strength of the joint was reduced gradually. In all mast all of the welding regimes, the tensile fracture took place at the tunnel defects (see Fig. 4(a)) except for the case of the joint fabricated at the welding speed of 70 mm/min (see Fig. 4(b)). In this case, the joint was fractured at the interface. Comparing the tensile strength of the joint fabricated at the welding speed of 100 mm/min with that at the welding speed of 70 mm/min, it can be seen that the interface bonding between copper and aluminum was reduced at the low welding speed (at 70 mm/min).

Table 1. Macrostructure of the FSW Cu/Al at various welding speeds.

Welding speed, mm/min.	Macrostructure	
200		
160		
130		
100		
70		

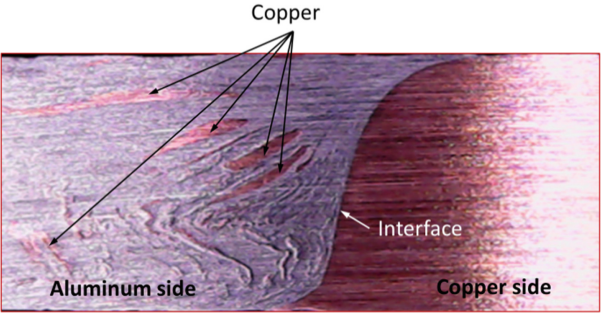


Fig. 2. The mixed copper into aluminum (welding speed of 200 mm/min).

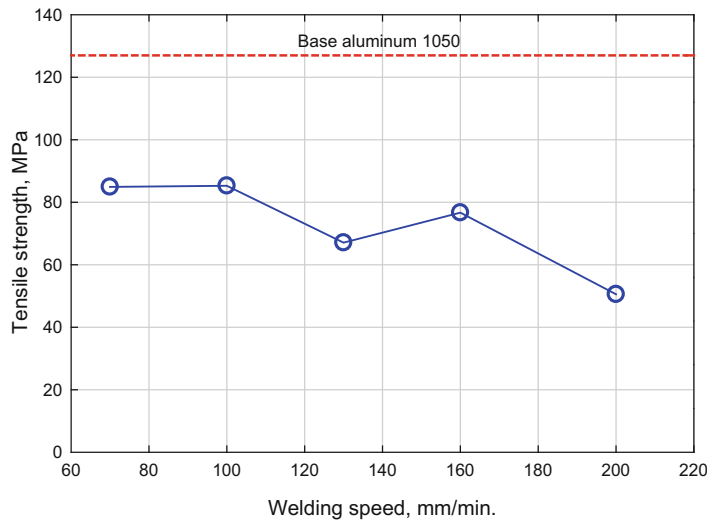


Fig. 3. Tensile strength of the joint at various welding speeds.

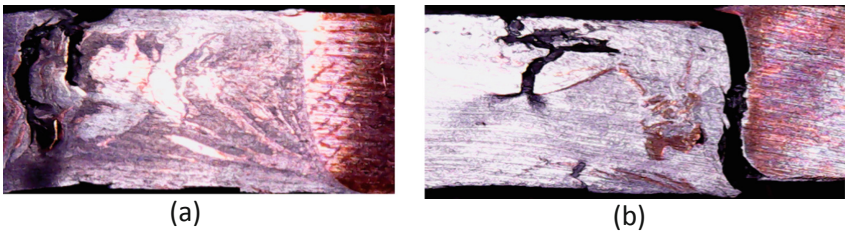


Fig. 4. The joint fracture (a) at the tunnel defect (welding speed of 200 mm/min), (b) at the interface (welding speed of 70 mm/min).

4 Conclusions

The bonding between copper C1100 and aluminum 1050 could be obtained up to 85 MPa. The interface bonding between copper and aluminum was free defects, but the tunnel defects in the aluminum side seemed to be unavoidable. On the one hand, the tensile strength of the joint was reduced with the increase of the welding speed. The interface bonding between copper and aluminum was reduced with the low welding speeds (might be below 100 mm/mm).

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